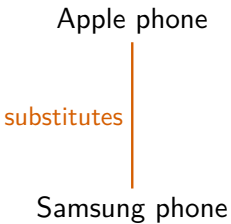
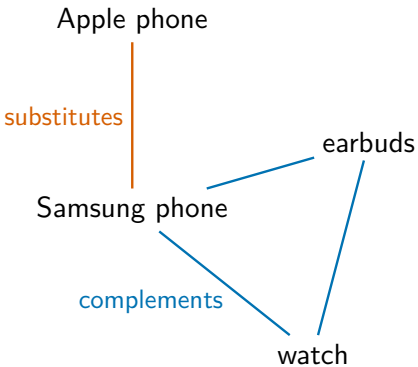
Robust Market Interventions

Andrea Galeotti (LBS)
Benjamin Golub (Northwestern)
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Eduard Talamàs (IESE)
Omer Tamuz (Caltech)

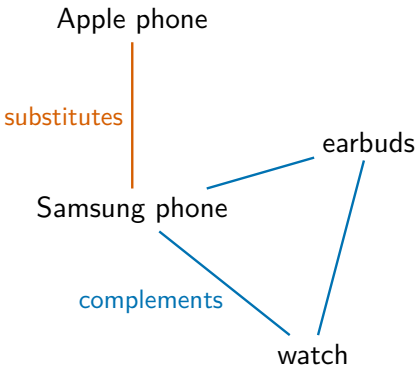
markets with many interacting goods



markets with many interacting goods

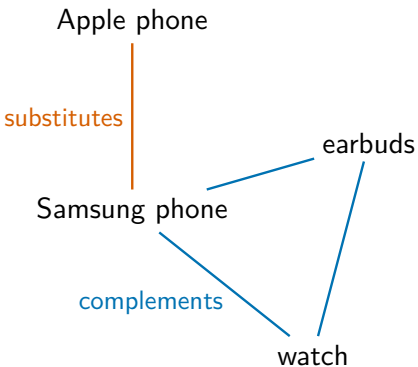


markets with many interacting goods



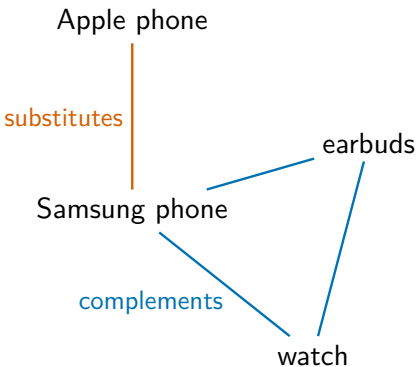
- spillovers across markets: lower phone prices increase demand for earbuds

markets with many interacting goods



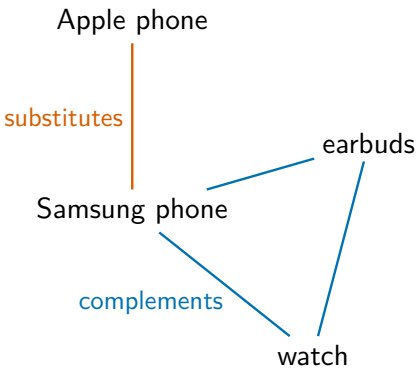
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- an authority (e.g., Amazon or a government) can tax or subsidize each good to raise total surplus

markets with many interacting goods



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- 256 goods imply 65,536 demand derivatives, each estimated with noise

markets with many interacting goods



- spillovers across markets: lower phone prices increase demand for earbuds
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when do noisy estimates permit surplus gains with high probability in every market?

the answer

robust surplus gains depend on **one market property**

significant structure:
roughly, large groups of complementary goods

the answer

robust surplus gains depend on **one market property**

significant structure:
roughly, large groups of complementary goods

with this property

- a rule using only noisy estimates reliably approaches the complete-information optimum
- each $1 spent yields about $2 of market surplus
- consumers essentially unaffected

the answer

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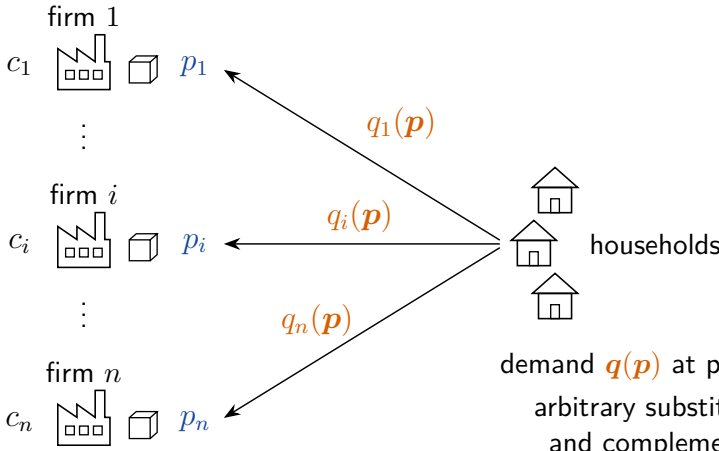
- in some markets, no rule guarantees even a net gain

significant structure ensures both *statistical* and *economic* properties,
and their interaction is interesting

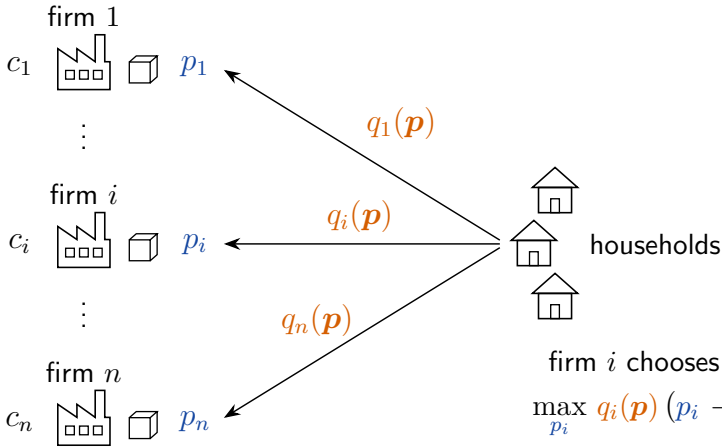
part I:

the oligopoly game

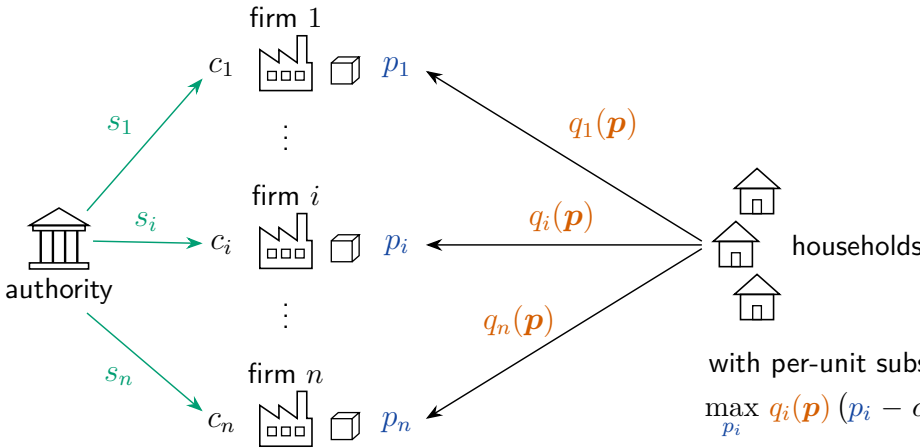
the game


firm 1
c_1
p_1
$q_1(\boldsymbol{p})$
firm i
c_i
p_i
$q_i(\boldsymbol{p})$
households
$q_n(\boldsymbol{p})$
firm n
c_n
p_n
demand $\boldsymbol{q}(\boldsymbol{p})$ at prices $\boldsymbol{p}$:
arbitrary substitutes
and complements

the game


firm 1
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p_i
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households
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firm n
c_n
p_n
firm i chooses p_i:
$\max_{p_i} q_i(\boldsymbol{p})\,(p_i - c_i)$
Nash equilibrium $(\boldsymbol{p}^0, \boldsymbol{q}^0)$:
status quo

the game



with per-unit subsidy s_i:

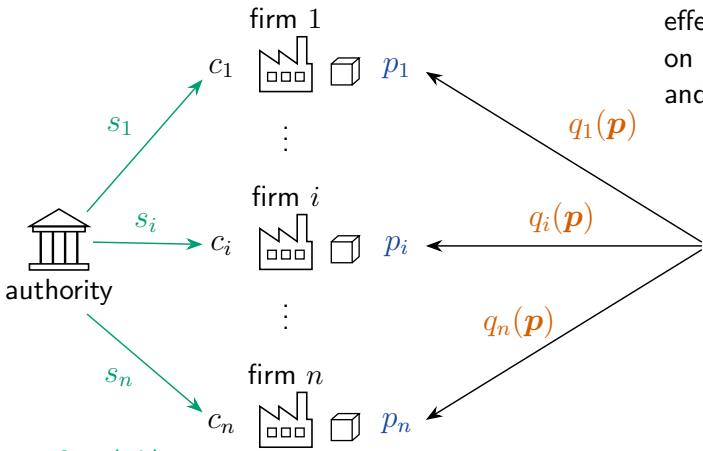
$$\max_{p_i} \; q_i(\boldsymbol{p}) \, (p_i - c_i + s_i)$$

effective marginal cost: $c_i - s_i$

$s_i > 0$: subsidy

$s_i < 0$: tax

the game



effect of a small subsidy $\boldsymbol{s}$
on equilibrium prices
and quantities

households

with per-unit subsidy s_i:

$$\max_{p_i} \; q_i(\boldsymbol{p}) \, (p_i - c_i + s_i)$$

effective marginal cost: $c_i - s_i$

$s_i > 0$: subsidy
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impact on equilibrium

demand responses at the status quo:

$$D_{ij} = \frac{\partial q_i}{\partial p_j}(\boldsymbol{p}^0)$$

demand matrix $\boldsymbol{D}$:
symmetric, negative semidefinite

$D_{ij} > 0$: **substitutes**
$D_{ij} < 0$: **complements**

units: $D_{ii} = -1$

the math →

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a small subsidy vector $\boldsymbol{s}$ changes equilibrium prices and quantities:

$$\Delta \boldsymbol{p} = -(\boldsymbol{I} - \boldsymbol{D})^{-1}\boldsymbol{s}, \quad \Delta \boldsymbol{q} = \boldsymbol{D}\,\Delta \boldsymbol{p}$$

$\Delta \boldsymbol{p}$, $\Delta \boldsymbol{q}$: first-order changes in prices and quantities

the math →

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assumption: near the status quo, each good's demand is linear in its own price

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the inverse incorporates cross-market spillovers: each of its entries depends on every entry of $\boldsymbol{D}$

assumption: near the status quo, each good's demand is linear in its own price

the math →

welfare

- **consumer surplus** CS: households' value of purchases minus expenditure
  $\mathrm{CS} = \sum_h \left[U^h(\boldsymbol{q}^h) - \boldsymbol{p} \cdot \boldsymbol{q}^h \right]$
- **producer surplus** PS: firms' profits, including subsidies
  $\mathrm{PS} = \sum_i \left(p_i - c_i + s_i \right) q_i$
- **spending** $S = \boldsymbol{s} \cdot \boldsymbol{q}$: subsidies times quantities sold
- **total surplus** $W = \mathrm{CS} + \mathrm{PS} - S$

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recall: impact on equilibrium

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$$\Delta \boldsymbol{q} = \boldsymbol{D}\, \Delta \boldsymbol{p} \qquad \text{quantities}$$

market power implies underproduction; at status-quo quantities $\boldsymbol{q}^0$,

$$\Delta W = \boldsymbol{q}^0 \cdot \Delta \boldsymbol{q} = \sum_i \text{markup}_i \, \Delta q_i$$

ΔW: first-order total-surplus gain; in our units, $\text{markup}_i = q_i^0$

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output gains are weighted by markups; price changes are transfers

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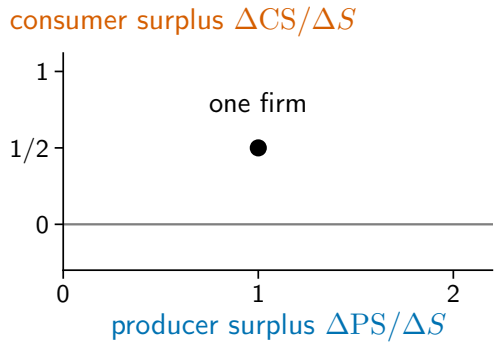
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goal: expand high-markup output

harder to achieve it: each subsidy moves every price and quantity

complete information

benchmark: one firm, linear demand



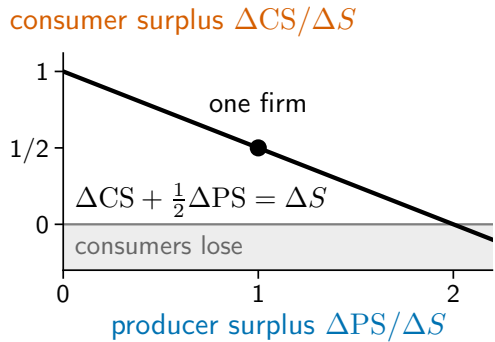
the math →

complete information

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Proposition 1: for every subsidy vector s, in every market,

$$\Delta\mathrm{CS} + \tfrac{1}{2}\,\Delta\mathrm{PS} = \Delta S$$



the math →

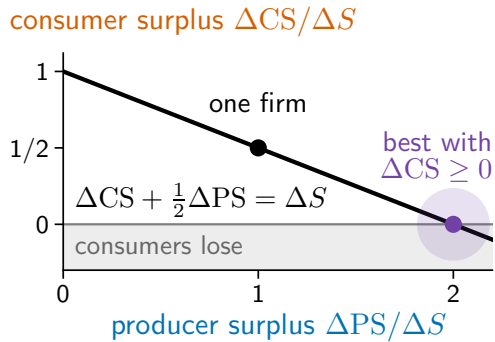
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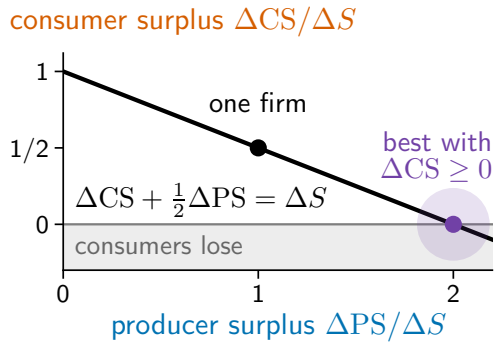
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with interacting goods, attaining **the corner** requires the full demand matrix $\boldsymbol{D}$



the math →

the problem

choose subsidies $\boldsymbol{s}$, with a fixed budget,
to make the *true* gain large:

$$\Delta W(\boldsymbol{s}) \;=\; \underbrace{(\boldsymbol{q}^0)^{\mathsf{T}}}_{\text{markups}} \underbrace{\left[-\boldsymbol{D}\,(\boldsymbol{I}-\boldsymbol{D})^{-1}\right]}_{f(\boldsymbol{D})} \boldsymbol{s}$$

$f(\boldsymbol{D})$: how quantities respond to $\boldsymbol{s}$

recall: where ΔW comes from

$\Delta \boldsymbol{p} = -(\boldsymbol{I}-\boldsymbol{D})^{-1}\boldsymbol{s}$	prices	$\Delta W = \boldsymbol{q}^0 \cdot \Delta \boldsymbol{q}$	total surplus
$\Delta \boldsymbol{q} = \boldsymbol{D}\,\Delta \boldsymbol{p}$	quantities	$\text{markup}_i = q_i^0$	in our units

the authority's problem

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the catch: $\boldsymbol{D}$ is unknown; the authority sees only a noisy estimate $\widehat{\boldsymbol{D}} = \boldsymbol{D} + \boldsymbol{E}$, with errors as large as own-price effects

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the tempting shortcut

plug in $\widehat{\boldsymbol{D}}$ for $\boldsymbol{D}$ and act as if the estimate were right:

$$\widehat{\Delta W}(\boldsymbol{s}) \;=\; \underbrace{(\widehat{\boldsymbol{q}}^0)^{\mathsf{T}}}_{\text{estimated}} \underbrace{\left[-\widehat{\boldsymbol{D}}\,(\boldsymbol{I}-\widehat{\boldsymbol{D}})^{-1}\right]}_{f(\widehat{\boldsymbol{D}})} \boldsymbol{s}$$

the same function, evaluated at $\widehat{\boldsymbol{D}}$

recall: where ΔW comes from

$\Delta \boldsymbol{p} = -(\boldsymbol{I}-\boldsymbol{D})^{-1}\boldsymbol{s}$	prices	$\Delta W = \boldsymbol{q}^0 \cdot \Delta \boldsymbol{q}$	total surplus
$\Delta \boldsymbol{q} = \boldsymbol{D}\,\Delta \boldsymbol{p}$	quantities	$\text{markup}_i = q_i^0$	in our units

the authority's problem

the problem

choose subsidies s, with a fixed budget, to make the *true* gain large:

$$\Delta W(s) \;=\; \underbrace{(q^0)^{\mathsf{T}}}_{\text{markups}} \underbrace{\left[-D\,(I-D)^{-1}\right]}_{f(D)}\, s$$

$f(D)$: how quantities respond to s

the catch: D is unknown; the authority sees only a noisy estimate $\widehat{D} = D + E$, with errors as large as own-price effects

the tempting shortcut

plug in $\widehat{D}$ for D and act as if the estimate were right:

$$\widehat{\Delta W}(s) \;=\; \underbrace{(\hat{q}^0)^{\mathsf{T}}}_{\text{estimated}} \underbrace{\left[-\widehat{D}\,(I-\widehat{D})^{-1}\right]}_{f(\widehat{D})}\, s$$

the same function, evaluated at $\widehat{D}$

but s is scored by the *true* ΔW, and $f(\widehat{D})$ can be far from $f(D)$: the inverse can even flip the sign

recall: where ΔW comes from

$\Delta p = -(I-D)^{-1}s$	prices	$\Delta W = q^0 \cdot \Delta q$	total surplus
$\Delta q = D\,\Delta p$	quantities	$\text{markup}_i = q_i^0$	in our units

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the same function, evaluated at $\widehat{\boldsymbol{D}}$

but $\boldsymbol{s}$ is scored by the *true* ΔW, and $f(\widehat{\boldsymbol{D}})$ can be far from $f(\boldsymbol{D})$: the inverse can even flip the sign

a simple **control problem**: choose $\boldsymbol{s}$ for the true $\boldsymbol{D}$,
but $\boldsymbol{D}$ is seen only through $\widehat{\boldsymbol{D}}$: we need a **robust** control

related literature: markets and networks

Market power across markets and pass-through

Azar and Vives (2021); Pellegrino (2021);
Ederer and Pellegrino (2021); Weyl and Fabinger (2013)

Cross-market inefficiencies and cost pass-through. We study small taxes and subsidies that increase total surplus when cross-price effects are estimated imprecisely.

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Interventions in network games

Ballester, Calvó-Armengol and Zenou (2006); Galeotti, Golub and Goyal (2020)

Existing targeting results assume known spillovers. We derive spectral pass-through in oligopoly and target demand directions that remain identifiable under noise.

related literature: inference and decisions

High-dimensional statistics

Davis and Kahan (1970); Chen, Chi, Fan and Ma (2021)

Spectral methods recover common structure in a noisy matrix. We recover demand directions with high welfare returns to subsidies, without estimating every interaction.

related literature: inference and decisions

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Statistical decision theory

Wald (1950); Manski (2004); Kitagawa and Tetenov (2018); Athey and Wager (2021)

We require a high-probability welfare guarantee in every market state, with probability over the noisy signal and no prior over states.

part II: the main result

robust surplus gains depend on **one market property**

with this property

- a rule using only noisy estimates reliably approaches the complete-information optimum
- each $1 spent yields about $2 of market surplus
- consumers essentially unaffected

without it

- in some markets, no rule guarantees even a net gain

the complete-information welfare bound:
$2 of market surplus per $1 spent, with consumers unharmed

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next: attaining this bound from noisy estimates;
then, the mechanism

the authority's information and goal

information

the authority observes noisy demand and quantity estimates

$$\widehat{\boldsymbol{D}} \;=\; \boldsymbol{D} + \boldsymbol{E}, \qquad \widehat{\boldsymbol{q}}^{0} \;=\; \boldsymbol{q}^{0} + \text{noise}$$

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$\boldsymbol{E}$ is symmetric, with independent mean-zero upper-triangular entries on the scale of own-price effects

a **market** is a pair $\theta = (\boldsymbol{D}, \boldsymbol{q}^{0})$, all that welfare depends on

the authority knows a set Θ of possible markets, with no prior

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an *intervention rule* maps the estimates $(\widehat{\boldsymbol{D}}, \widehat{\boldsymbol{q}}^{0})$ to a subsidy vector $\boldsymbol{s}$; it achieves a property ε-**robustly** if

$$\mathbb{P}_{\theta}[\text{property}] \;\geq\; 1 - \varepsilon \quad \text{for every } \theta \in \Theta$$

(such guarantees also hold in expectation)

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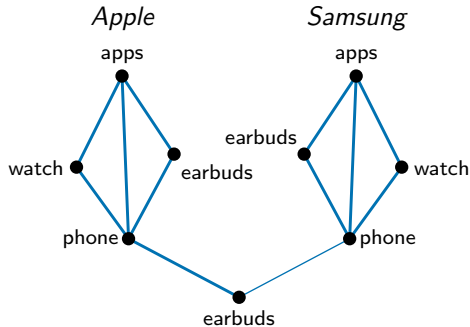
(such guarantees also hold in expectation)

target: the corner of part I, robustly; consumer and producer gains satisfy

$$\Delta\text{CS} + \Delta\text{PS} \;\approx\; 2\,\Delta S, \qquad \Delta\text{CS} \approx 0$$

a dominant principal component of demand

a special case: **two ecosystems**
e.g. Apple and Samsung, each with its own phones, watches, earbuds and apps

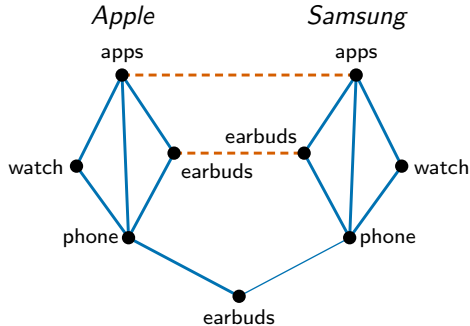


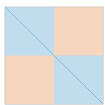
many such goods, $n = 256$

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complements within an ecosystem,
substitutes across





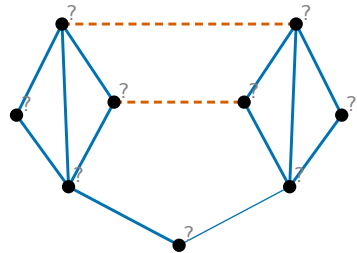
as a demand matrix,
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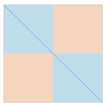
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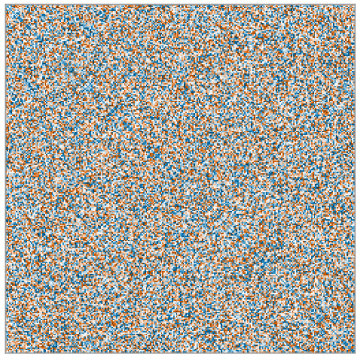
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the data: a noisy 256×256 demand matrix, goods in no particular order

can we recover anything meaningful? what?



observed $\widehat{\boldsymbol{D}}$, arbitrary order

the result for two ecosystems

n goods in two ecosystems;
which good is in which is unknown

within an ecosystem, complements
with cross-price effect c;
across, substitutes

demand estimates with
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complementarity may weaken as the market grows, but more slowly than $1/\sqrt{n}$

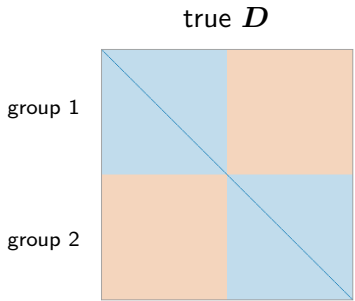
if instead $c\sqrt{n} \to 0$, the method doesn't work (formal statement)

a short film:

recovering structure from noisy data

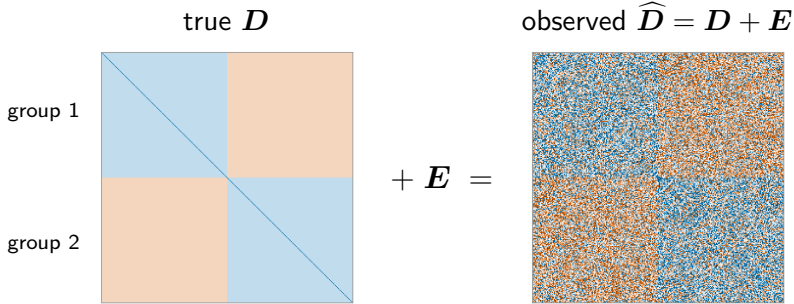
play the film the film as slides after the film →

a planted block structure



negative within groups, positive across; each entry about 0.3 in magnitude

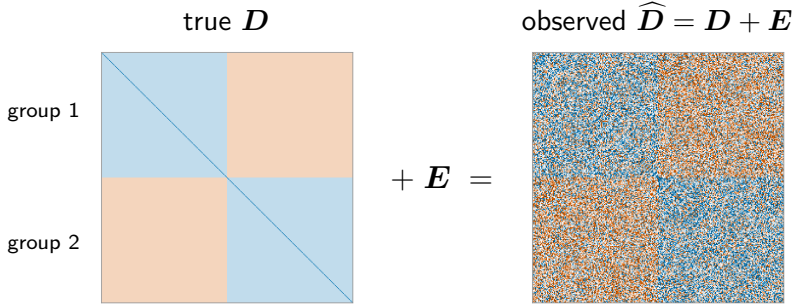
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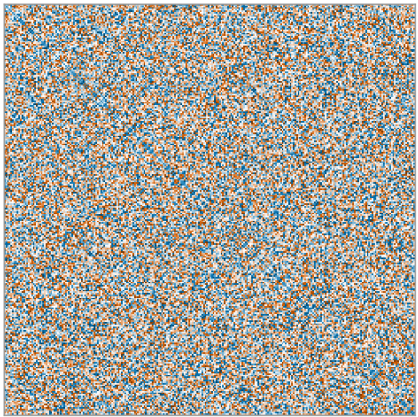


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with the goods in the right order, the blocks are faintly visible

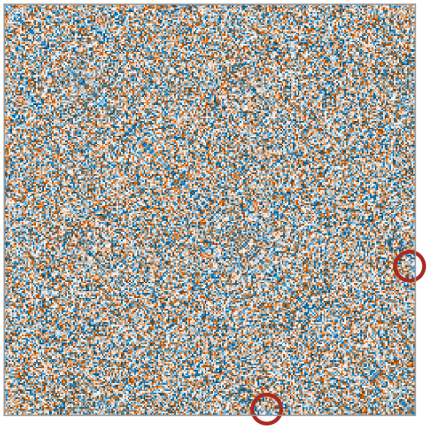
without the labels



observed $\widehat{\boldsymbol{D}}$, arbitrary order

the grouping that makes the picture legible is not known; under an arbitrary ordering of the goods, the pattern is no longer visible

without the labels

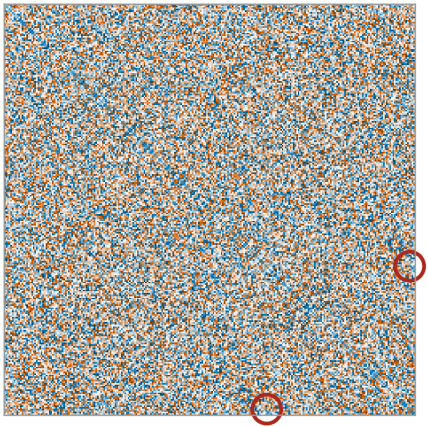


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the sign of an observed entry agrees with the sign of the true entry only **62%** of the time

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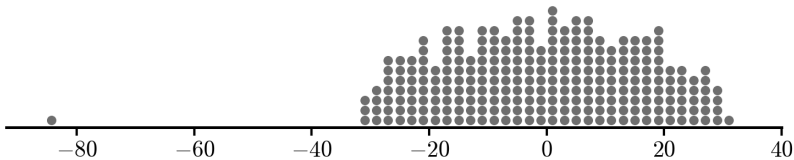
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can we recover the groups, or anything at all?

the spectrum of the observed matrix

compute the eigendecomposition of $\widehat{\boldsymbol{D}}$; in MATLAB, one line:

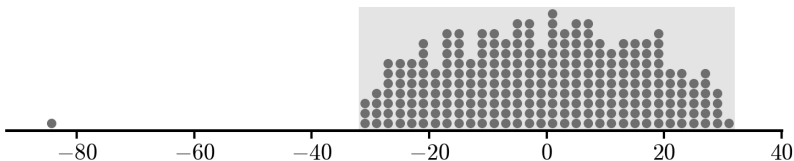
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>> [V, Lam] = eig(Dhat);
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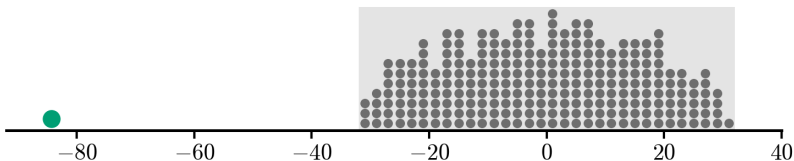


- most of the eigenvalues lie in a **band** around -1, of half-width of order $\sqrt{n}$, set by the noise (the centre is a feature of this example)

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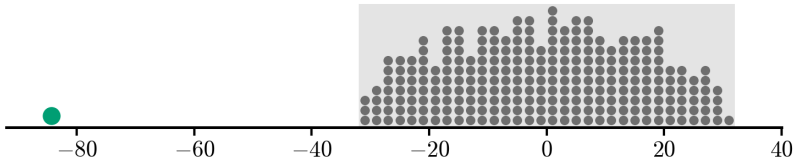


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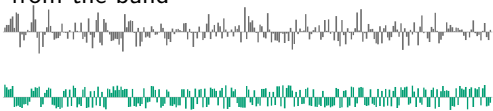
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- one eigenvalue, $\hat{\lambda} = -84$, lies far outside the band
- each eigenvalue has an eigenvector, one entry per good; the outlier's is the so-called **leading eigenvector** $\hat{\boldsymbol{u}}$

the leading eigenvector

from the band

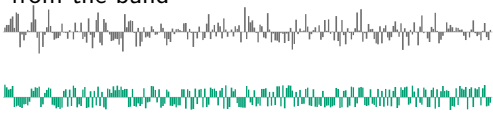


leading, $\hat{\boldsymbol{u}}$

entries in the arbitrary order:
neither shows structure

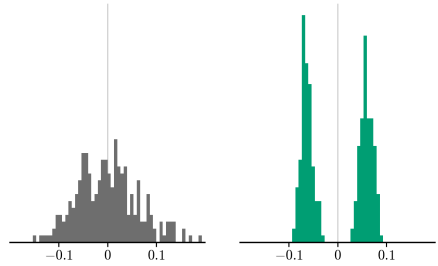
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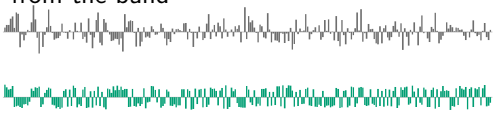


from the band leading

ignore the order and look at the distribution of entries: from the band, concentrated around zero

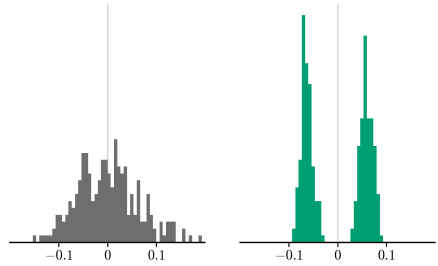
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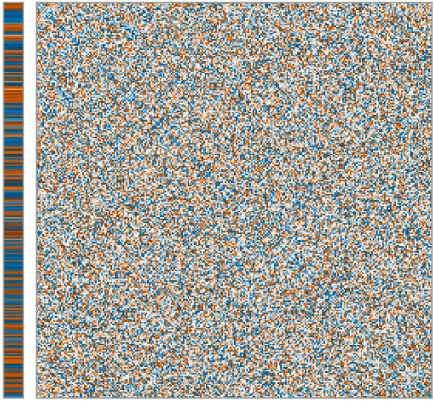
from the band leading

ignore the order and look at the distribution of entries: from the band, concentrated around zero

leading: **two clusters**, one positive and one negative, separated by a gap

two clusters because this example has two groups

assigning goods by sign

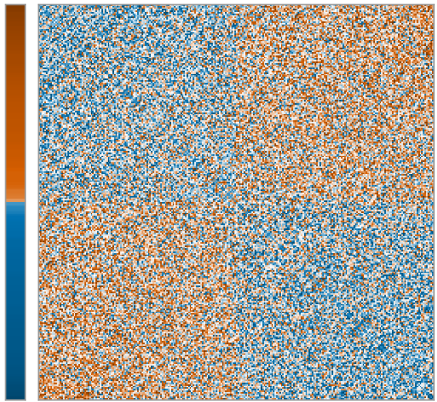


observed $\widehat{\boldsymbol{D}}$, with $\hat{\boldsymbol{u}}$ along the margin

the sign of each entry of $\hat{\boldsymbol{u}}$ can be used to assign its good to a group

the assignment uses only the observed matrix

assigning goods by sign



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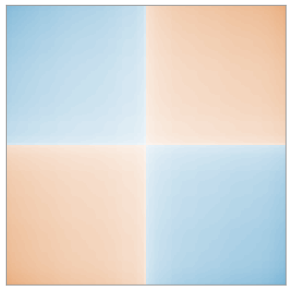
the assignment uses only the observed matrix

ordering the goods by this assignment restores the block structure

all 256 goods are assigned to their true groups

the rank-one term

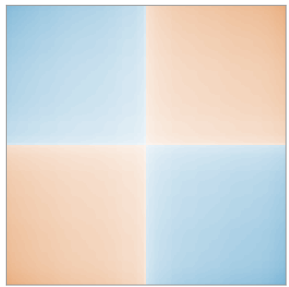
$\hat{\lambda}\,\hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^\top$, from $\widehat{\boldsymbol{D}}$



multiply $\hat{\boldsymbol{u}}$ by itself and scale by $\hat{\lambda} = -84$:
the same two blocks, negative within groups, positive across

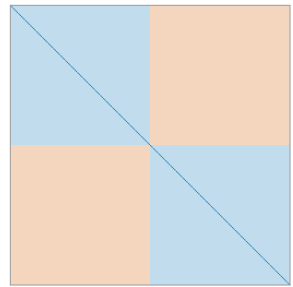
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vs.

true $\boldsymbol{D}$



multiply $\hat{\boldsymbol{u}}$ by itself and scale by $\hat{\lambda} = -84$:
the same two blocks, negative within groups, positive across

the planted structure, estimated from the observed matrix alone

why: the spectral theorem

a symmetric matrix is the sum of rank-one terms, one per eigenvector

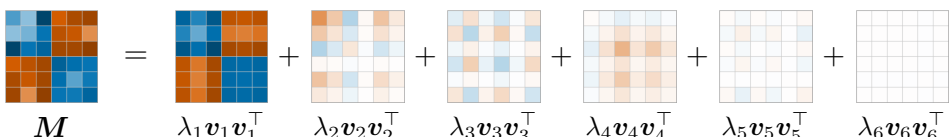
$$\boldsymbol{M} \;=\; \lambda_1 \boldsymbol{v}_1 \boldsymbol{v}_1^\top + \lambda_2 \boldsymbol{v}_2 \boldsymbol{v}_2^\top + \cdots + \lambda_n \boldsymbol{v}_n \boldsymbol{v}_n^\top$$

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six-by-six example: two groups of three, plus noise

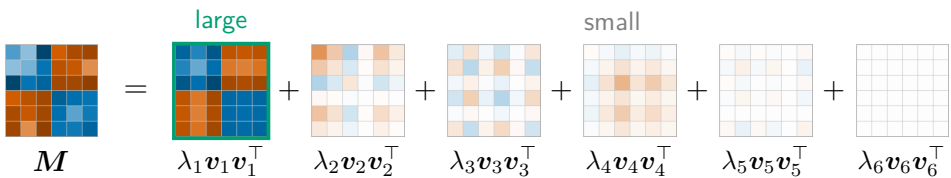


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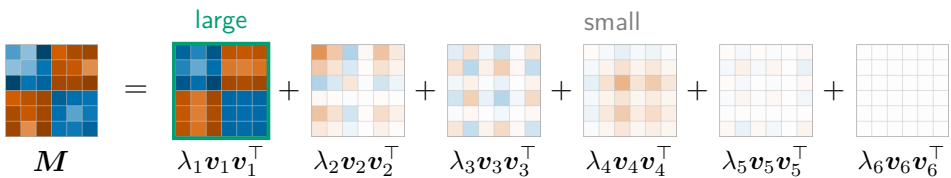
one block makes a much larger contribution, and it alone carries the two groups

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six-by-six example: two groups of three, plus noise



one block makes a much larger contribution, and it alone carries the two groups

a few much larger terms: random noise cannot change them much

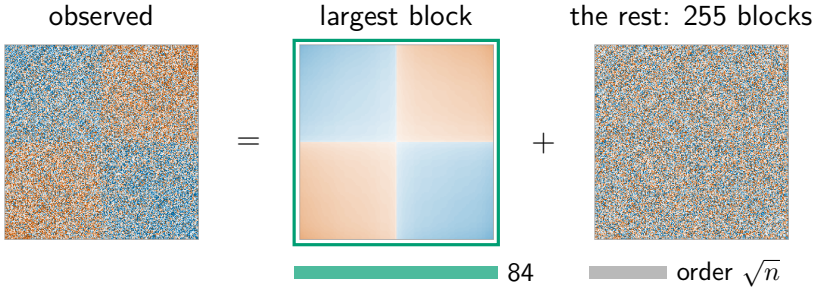
the largest block and the rest

observed | largest block | the rest: 255 blocks



back to the 256 goods: the observed matrix is its largest block plus the rest

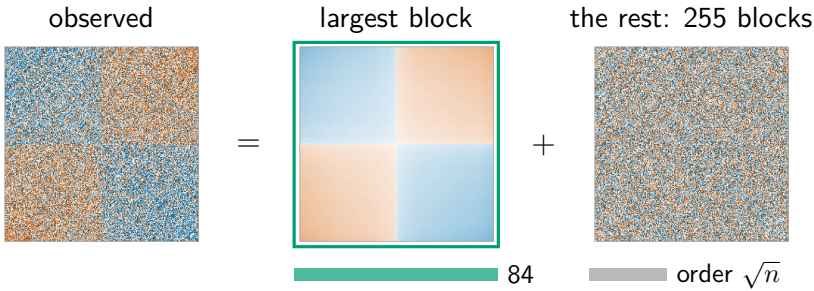
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back to the 256 goods: the observed matrix is its largest block plus the rest

largest block: eigenvalue -84; the rest, with all the noise: size of order $\sqrt{n}$

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the largest block alone carries the two groups

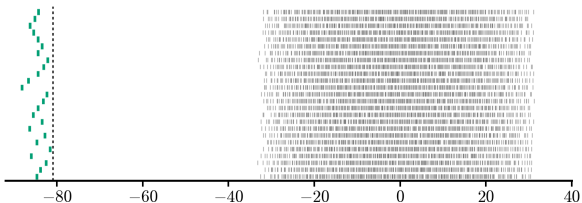
the noise redrawn

the noise changes the largest block in the spectral decomposition of the observed matrix, but only by a bounded amount

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hold $\boldsymbol{D}$ fixed and draw the noise again, 25 times:

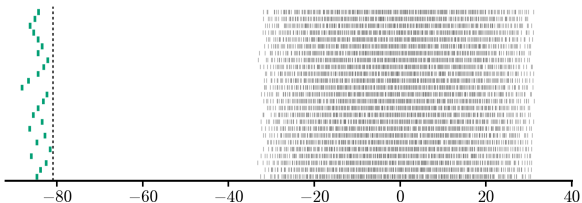


the band rearranges and its eigenvectors change completely;
the outlier moves a few units; the leading eigenvector barely changes

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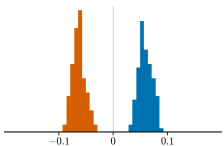
in every draw, no good is misassigned

how strong must the structure be?

reduce the planted eigenvalue, holding the noise fixed

entries of $\hat{u}$; color = true group, not used by the procedure

$|\lambda_1| = 81$



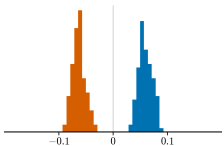
misassigned: 0

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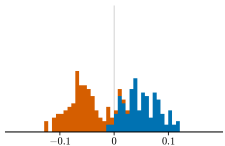
entries of $\hat{u}$; color = true group, not used by the procedure

$|\lambda_1| = 81$



misassigned: 0

$|\lambda_1| = 33$



misassigned: 9

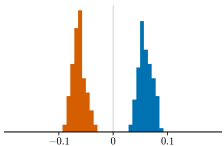
at 33, nine goods are misassigned

how strong must the structure be?

reduce the planted eigenvalue, holding the noise fixed

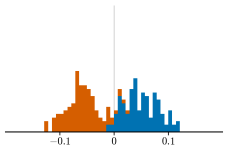
entries of $\hat{u}$; color = true group, not used by the procedure

$|\lambda_1| = 81$



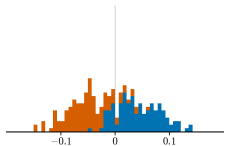
misassigned: 0

$|\lambda_1| = 33$



misassigned: 9

$|\lambda_1| = 25$



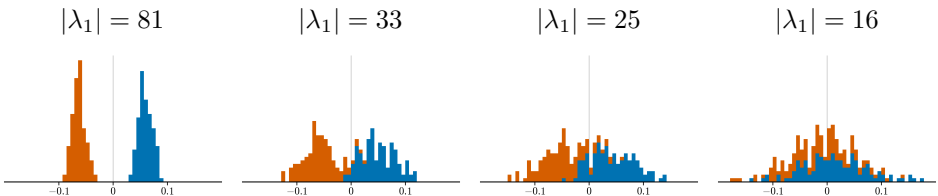
misassigned: 35

at 25, the clusters begin to merge

how strong must the structure be?

reduce the planted eigenvalue, holding the noise fixed

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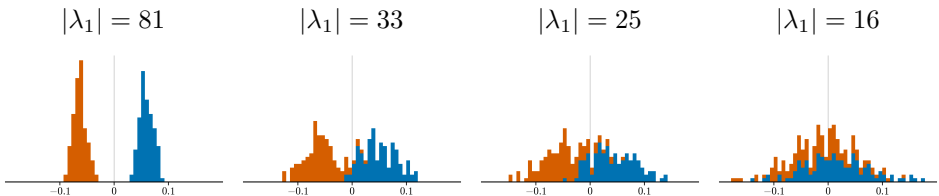


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misassigned: 0 | misassigned: 9 | misassigned: 35 | misassigned: 104

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structure weaker than the noise is not recovered; structure sufficiently stronger is

back to demand

nothing here uses the economics; it is a property of symmetric matrices

a planted component that is large relative to the rest is recovered by its eigenvector

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back to demand

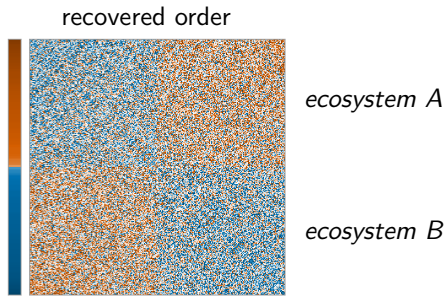
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in the model, the planted component is the **ecosystem structure** of demand

the leading eigenvector is what the intervention rule uses

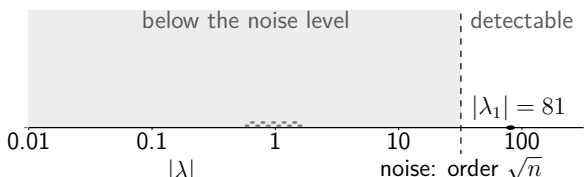


significant structure

significant structure requires detectability and quantity relevance:

1. some eigenvalues are far larger than the noise level, of order $\sqrt{n}$

$$|\lambda| \geq b(n), \quad \|\boldsymbol{E}\|/b(n) \to 0$$



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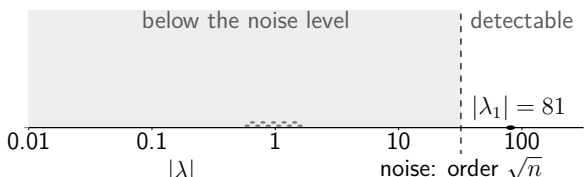
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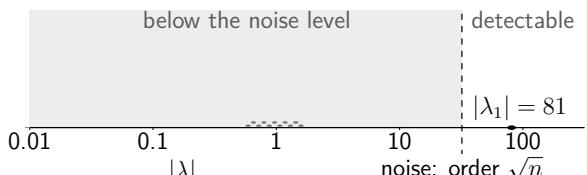
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economic interpretation: large-scale complementarity

also: quantity noise no larger than quantities

main result

Theorem 1. Under significant structure, for any $\varepsilon > 0$ and budget $\bar{S}$, **there exists a rule** that ε-robustly achieves, in large markets:

the math →

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$$\Delta\mathrm{CS} + \Delta\mathrm{PS} \;\geq\; (2 - \varepsilon)\,\Delta S$$

the math →

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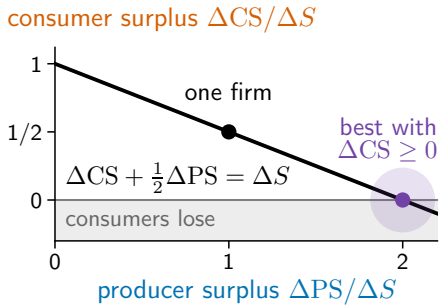
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the complete-information **corner**

first-order effects; regularity: backup

the math →

the rule

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2. project observed quantities $\widehat{\boldsymbol{q}}^0$ onto $\widehat{\mathcal{L}}$
3. scale the subsidy vector to budget $\bar{S}$

$$\boldsymbol{s} \;=\; \bar{S}\,\frac{\mathbb{P}_{\widehat{\mathcal{L}}}\,\widehat{\boldsymbol{q}}^0}{\|\mathbb{P}_{\widehat{\mathcal{L}}}\,\widehat{\boldsymbol{q}}^0\|^2}$$

no inversion of $\widehat{\boldsymbol{D}}$

projection averages over many noisy entries

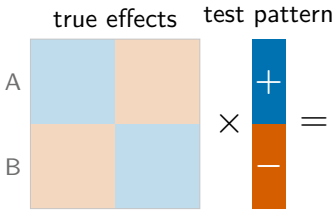
next: intuition and proof

the math →

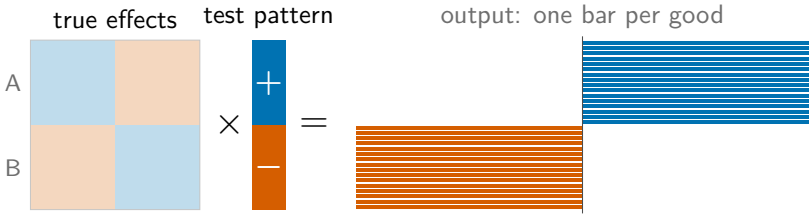
part III:

economic intuition

structure and noise



structure and noise

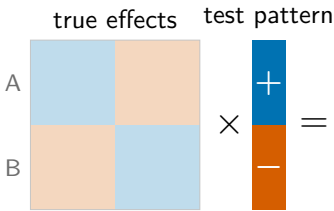


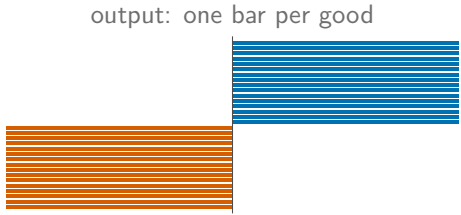
adds up

each term $+0.32$

256 terms: about 81

structure and noise

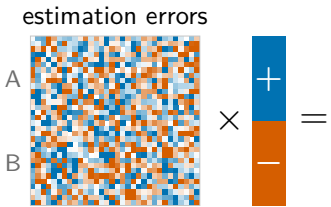




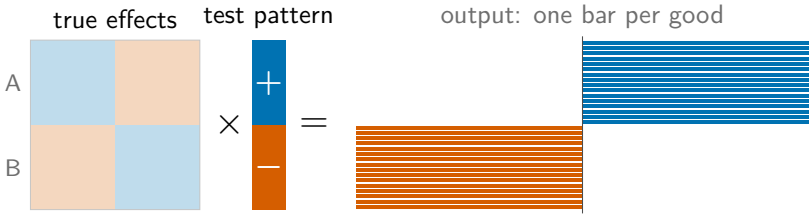
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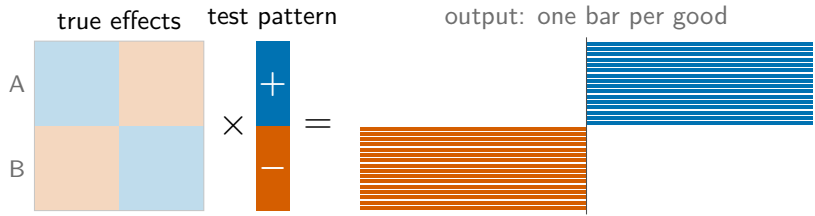
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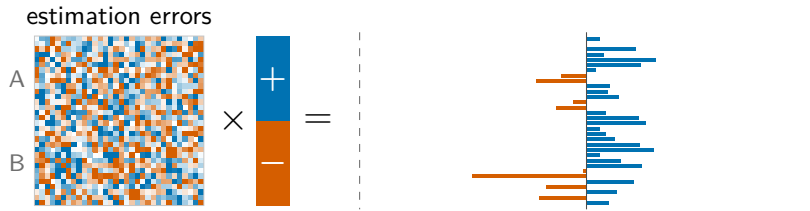
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with n goods, structure grows like n, noise only like $\sqrt{n}$

spectral price theory

demand, written in its eigenvectors (the spectral theorem, as in the film):

$$\boldsymbol{D} \;=\; \lambda_1 \boldsymbol{u}^1 (\boldsymbol{u}^1)^\mathsf{T} + \lambda_2 \boldsymbol{u}^2 (\boldsymbol{u}^2)^\mathsf{T} + \cdots + \lambda_n \boldsymbol{u}^n (\boldsymbol{u}^n)^\mathsf{T}$$

each eigenvector $\boldsymbol{u}^\ell$ is a **bundle** of goods; $|\lambda_\ell|$: how strongly its sales respond to its price

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one good alone

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quantity $\boldsymbol{u}^\ell \cdot \Delta \boldsymbol{q} = \dfrac{|\lambda_\ell|}{1+|\lambda_\ell|}\, \boldsymbol{u}^\ell \cdot \boldsymbol{s}$

welfare $\Delta W = \displaystyle\sum_\ell \frac{|\lambda_\ell|}{1+|\lambda_\ell|}\, \Delta S_\ell$

$\Delta S_\ell = (\boldsymbol{u}^\ell \cdot \boldsymbol{q}^0)(\boldsymbol{u}^\ell \cdot \boldsymbol{s})$, spending on bundle ℓ

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the math →

subsidy incidence

per $1 of subsidy	**isolated good** $\lvert\lambda\rvert = 1$
price reduction	 50¢
consumer surplus	 50¢
producer surplus	 $1.00
market surplus consumers + producers	 $1.50

subsidy incidence

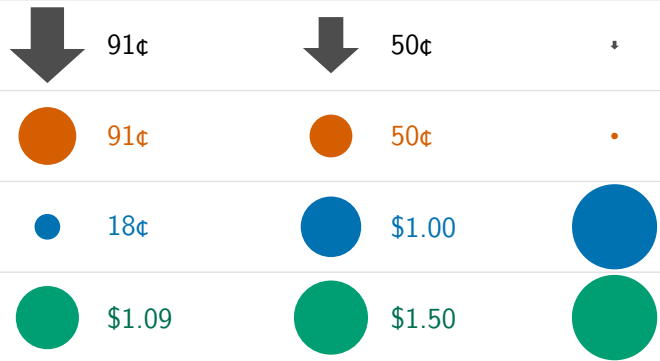
per $1 of subsidy	**weak demand response** $\lvert\lambda\rvert = 0.1$	**isolated good** $\lvert\lambda\rvert = 1$
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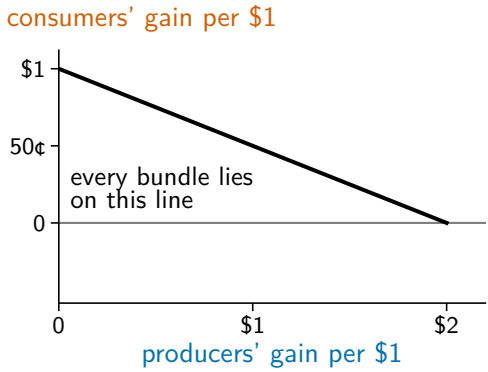


little pass-through, output expands: about **$2 of market surplus** per $1

surplus, eigenvector by eigenvector

any subsidy is a mix of bundles; its gain per \$1 averages the bundles' gains, weighted by spending

each bundle lands on the line of Proposition 1; its eigenvalue says where



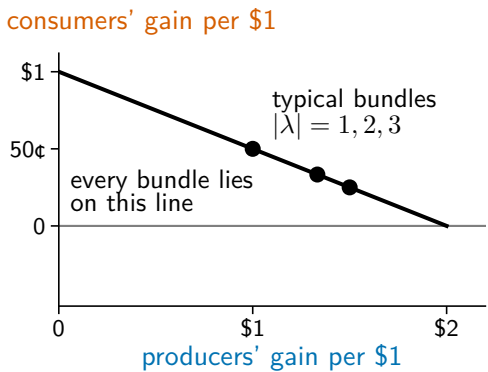
the math →

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the math →

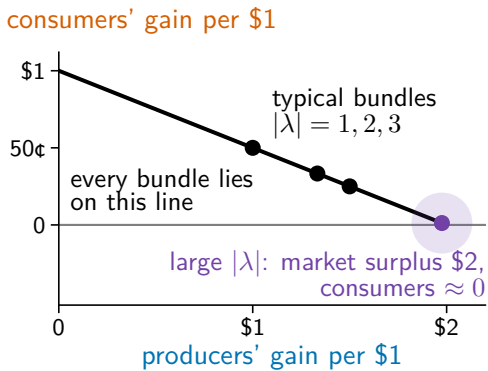
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about \$2; consumers about 0: the corner



the math →

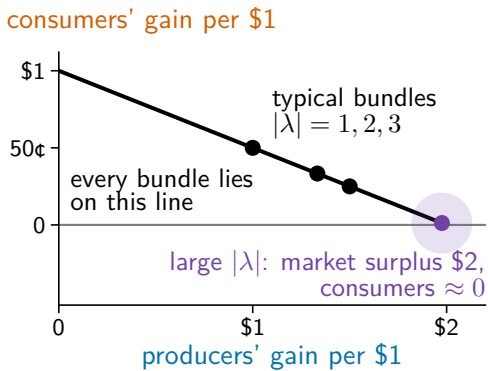
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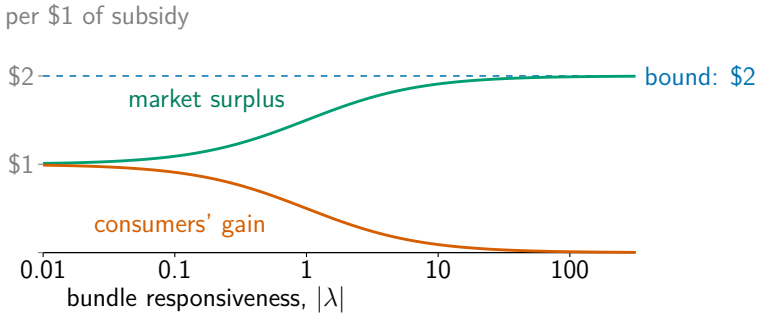
bundles with **large** $|\lambda|$:
about \$2; consumers about 0: the corner



to reach the corner, spend only on bundles with large $|\lambda|$, if we can find them

the math →

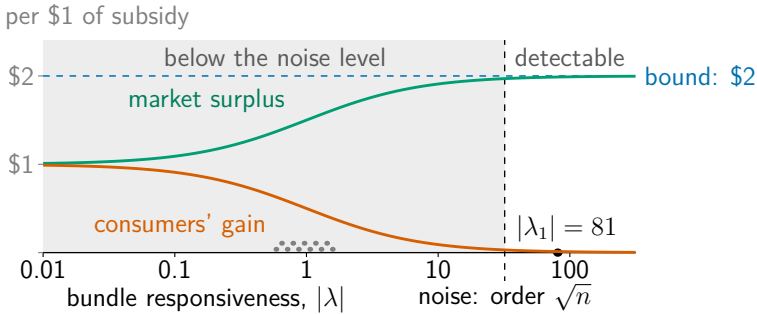
recovery and surplus



surplus: per $1, strong patterns yield about $2 of market surplus, consumers about 0

the math →

recovery and surplus

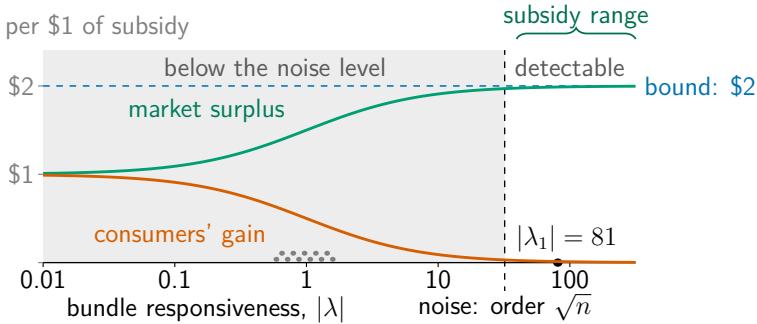


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recovery: only strong patterns stand out from the noise

the math →

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recoverable directions yield the largest subsidy gains

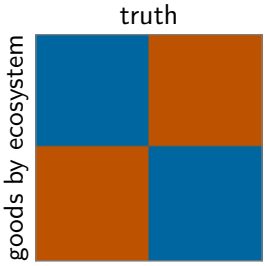
the math →

part III, continued:

formal argument

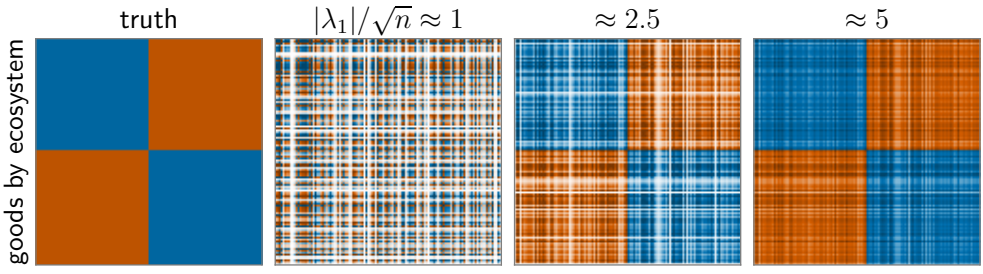
Weyl and Davis–Kahan (reserve) $\rightarrow$

recovering structure



blue: complements vermillion: substitutes

recovering structure



estimated matrix $-\widehat{\boldsymbol{u}}^1(\widehat{\boldsymbol{u}}^1)^\top$ as $|\lambda_1|$ increases:

noise $\rightarrow$ partial recovery $\rightarrow$ distinct groups

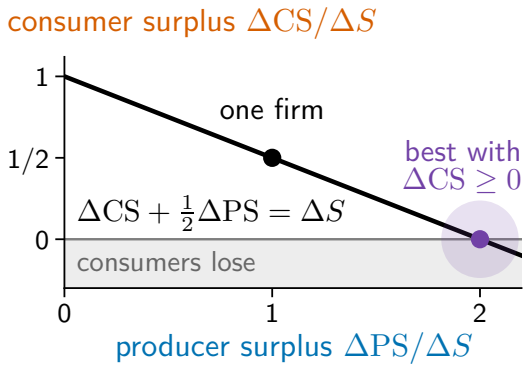
complementary groups are recovered without estimating every entry

surplus by eigenvector

Lemma. A \$1 subsidy along a demand eigenvector with eigenvalue λ affects only that direction:

consumer surplus $\dfrac{1}{1+|\lambda|}$

market surplus $\dfrac{1+2|\lambda|}{1+|\lambda|}$



the math →

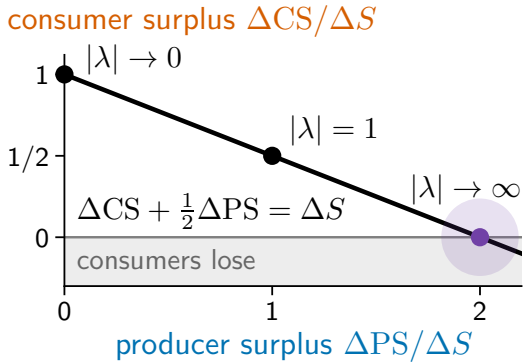
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a 1¢ price cut increases sales by $|\lambda|$ units; large $|\lambda|$ implies low pass-through and greater output



the math →

surplus by eigenvector

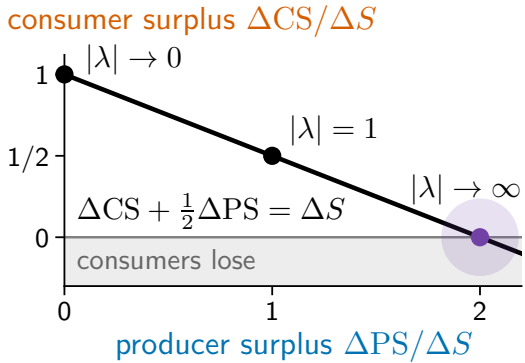
Lemma. A \$1 subsidy along a demand eigenvector with eigenvalue λ affects only that direction:

consumer surplus $\dfrac{1}{1+|\lambda|}$

market surplus $\dfrac{1+2|\lambda|}{1+|\lambda|}$

a 1¢ price cut increases sales by $|\lambda|$ units; large $|\lambda|$ implies low pass-through and greater output

large $|\lambda|$ suffices for near-maximal surplus



the math →

proof outline

1. the estimated high-response span $\widehat{\mathcal{L}}$ approximates $\mathcal{L}$ (Davis–Kahan)

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Theorem 1 follows

the rule in the experiment

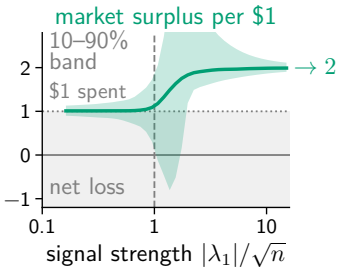
$$\boldsymbol{D} \;=\; -|\lambda_1|\,\boldsymbol{u}^1(\boldsymbol{u}^1)^\top \;+\; \underbrace{\text{aggregate}}_{|\lambda|=0.01} \;+\; \underbrace{\text{bulk}}_{|\lambda|\approx 1}$$

256 goods, two ecosystems, 500 noise draws per $|\lambda_1|$

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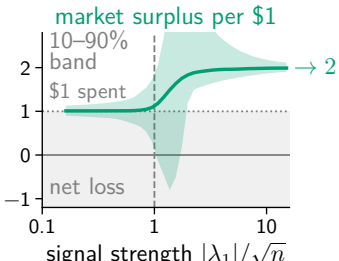
below the threshold $\sqrt{n}$

market surplus near the $1 spent, often below it

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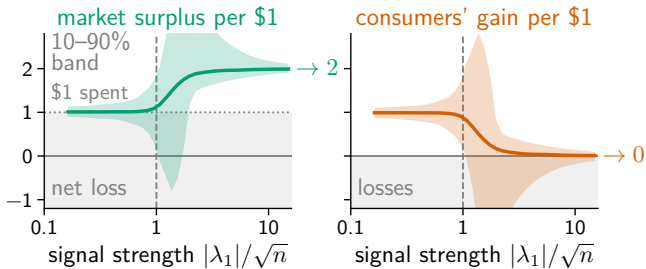
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the estimated eigenvector is random, so the subsidy falls on the aggregate bundle ($|\lambda| = 0.01$), lowering prices

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above the threshold

market surplus → $2

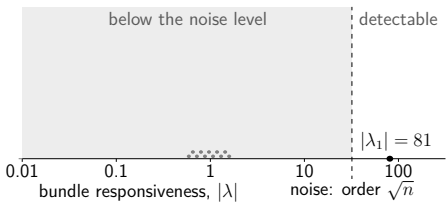
consumers' gain → 0

the limits of robust interventions

Proposition 3, for rules with $\Delta S > 0$:
1. in some markets **without** significant structure, no rule ε-robustly achieves

$$\Delta\mathrm{CS} + \Delta\mathrm{PS} \geq (1+\varepsilon)\,\Delta S$$

no robust net gain



the math →

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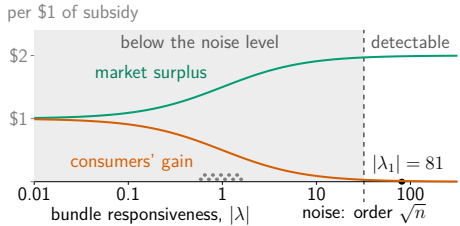
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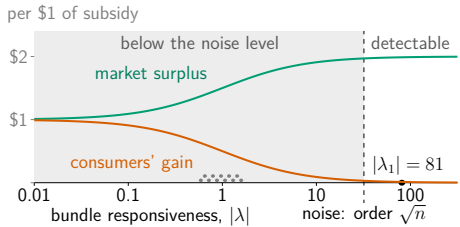
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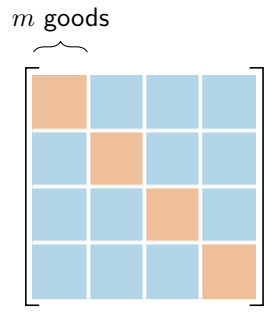


these limits follow from noise at small eigenvalues

the math →

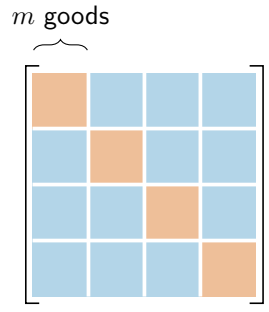
sources of significant structure

- **large-scale complementarity**: within ecosystems (our example) or across categories (rackets and shoes)



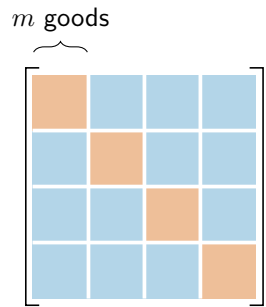
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- all-substitute (hedonic) demand models lack significant structure Pellegrino (2021)



substitutes within

complements across

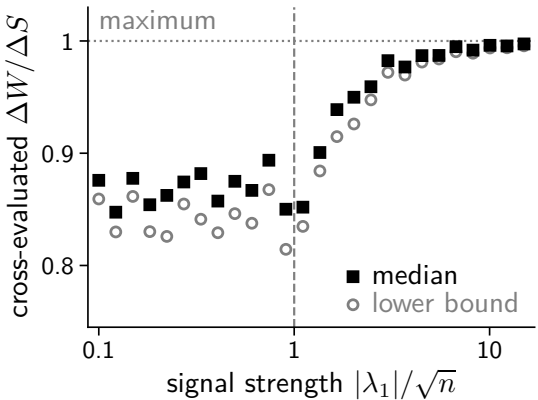
K categories of m goods

a diagnostic for significant structure

- estimate the rule on one data sample
- evaluate its net gain per dollar, $\Delta W/\Delta S$, on an independent sample

a diagnostic for significant structure

- estimate the rule on one data sample
- evaluate its net gain per dollar, $\Delta W/\Delta S$, on an independent sample
- significant structure $\Rightarrow$ ratio ≈ 1
- one-sided: success certifies; failure is inconclusive



extension: nonlinear demand

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needs only differentiable demand;
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two paths:

- direct oligopoly theory: recover the curvature too
- black-box pass-through: **compressed sensing**

other extensions: network games

our pricing game is a network game with linear best replies:
the network is the complements and substitutes among goods

other games with noisily known spillovers:

- public goods: align private and social returns
- teams under moral hazard: perturb contracts
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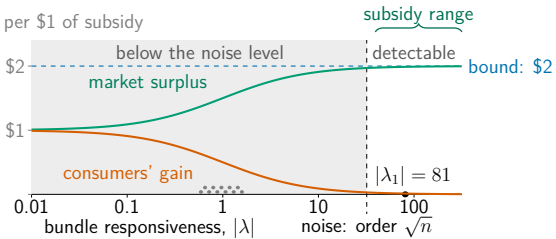
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we used consumer demand theory in a non-trivial way:
the Slutsky matrix is symmetric and negative semidefinite
not a straightforward extension, but a very interesting one

recap



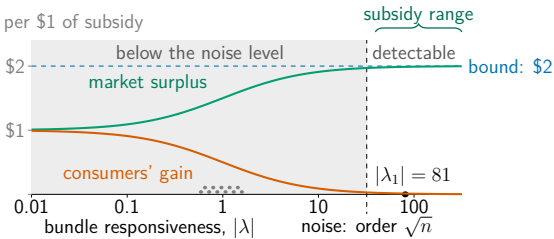
bound: the corner, \$2 of market surplus per \$1, consumers unharmed (Prop. 1)

recovery: $|\lambda| \lesssim \|\boldsymbol{E}\|$ is unresolved; above it, recover the top eigenspace (Weyl, Davis–Kahan)

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recap



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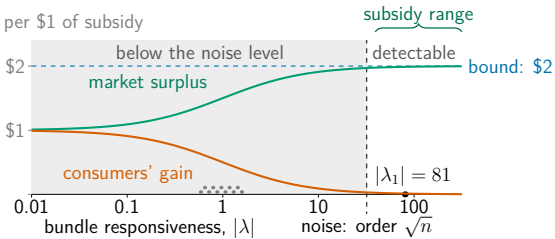
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recoverable eigendirections yield the largest subsidy gains

conclusion

standard IO

consistent estimation
in small markets

large markets

economically important, and large enough
for statistics to work in the large

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few tools for large-scale structure with non-trivial oligopolistic effects;
here, a theory of it, in contact with frontier statistics

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other applications, e.g. the macroeconomics of networks:
noisy measurements of individual links, but structure that transcends them

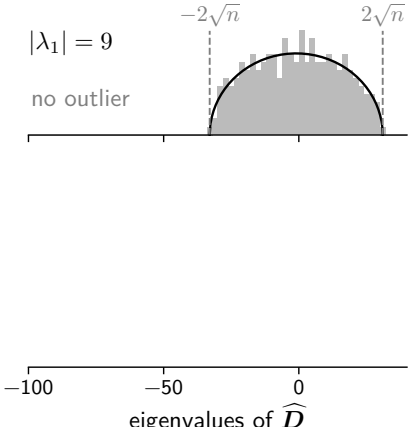
reserve:

the formal argument

play the film

back to part III →

detecting large eigenvalues



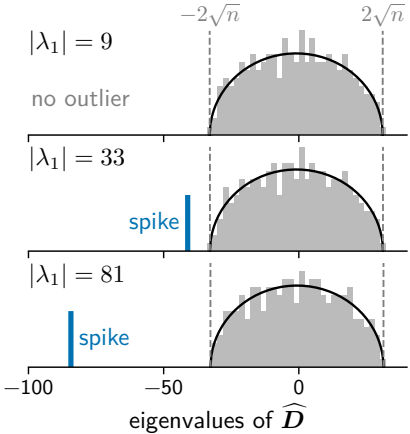
with n goods and noise matrix $\boldsymbol{E}$:

structure	$\lvert\lambda_1\rvert \approx 0.32\,n$	alignment
noise	$\lVert\boldsymbol{E}\rVert \approx 2\sqrt{n}$	signs cancel

with 256 goods: 81 versus a noise level of order $\sqrt{n}$

the math →

detecting large eigenvalues



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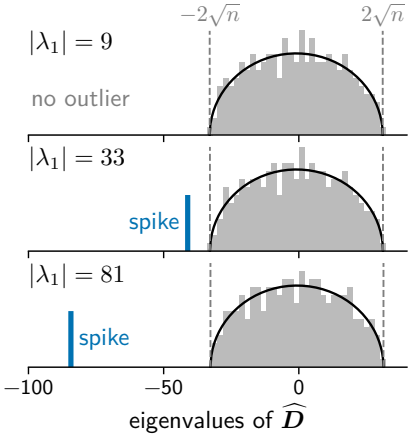
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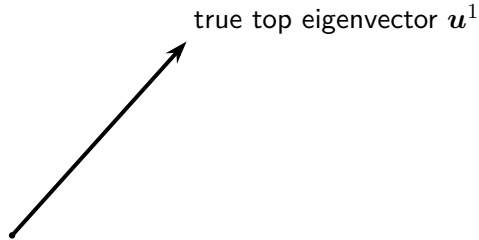
large eigenvalues are detectable

the math →

true and estimated eigenvectors

true demand matrix $\boldsymbol{D}$

$$\boldsymbol{D}\boldsymbol{u}^1 = \lambda_1 \boldsymbol{u}^1$$



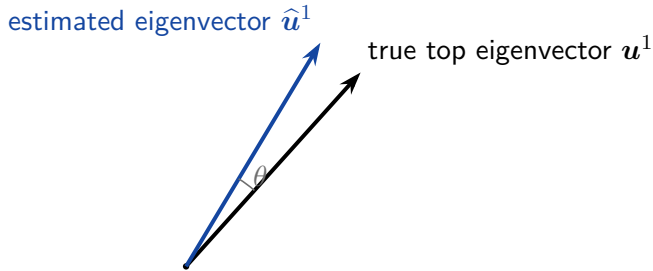
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estimated matrix $\widehat{\boldsymbol{D}} = \boldsymbol{D} + \boldsymbol{E}$

$$\widehat{\boldsymbol{D}}\widehat{\boldsymbol{u}}^1 = \widehat{\lambda}_1 \widehat{\boldsymbol{u}}^1$$



the estimated eigenvector approximates the true direction

eigenvector recovery

Theorem (Davis–Kahan 1970). With spectral separation gap, the rotation angle satisfies:

$$\theta \lesssim \frac{\|\boldsymbol{E}\|}{\mathrm{gap}}$$

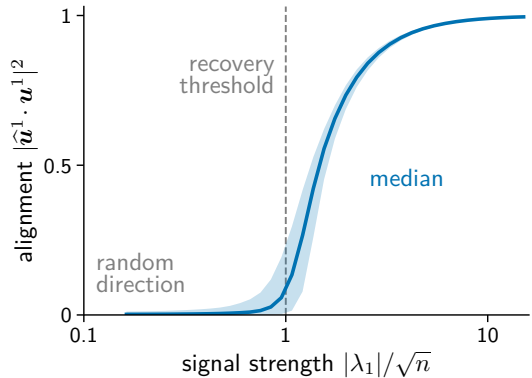
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simulated recovery above a threshold of order $\sqrt{n}$

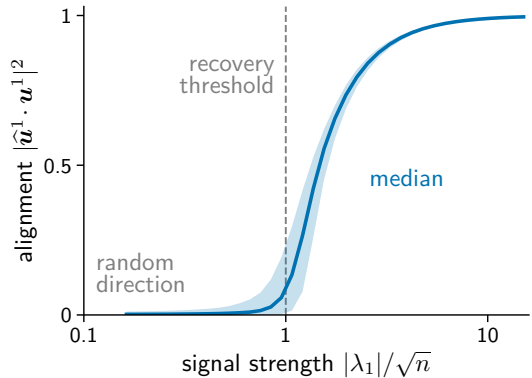
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with several large eigenvalues, their eigenspace is recovered



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Weyl's inequality

adding a symmetric matrix of norm $\|\boldsymbol{E}\|$
moves each eigenvalue by at most $\|\boldsymbol{E}\|$

$$|\lambda_k(\widehat{\boldsymbol{D}}) - \lambda_k(\boldsymbol{D})| \;\leq\; \|\boldsymbol{E}\|$$

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Davis–Kahan theorem

the rotation of the eigenvector is at most about the noise divided by the eigengap

$$\sin \angle(\hat{\boldsymbol{u}}, \boldsymbol{u}) \;\lesssim\; \frac{\|\boldsymbol{E}\|}{\mathsf{gap}}$$

here, about 24°

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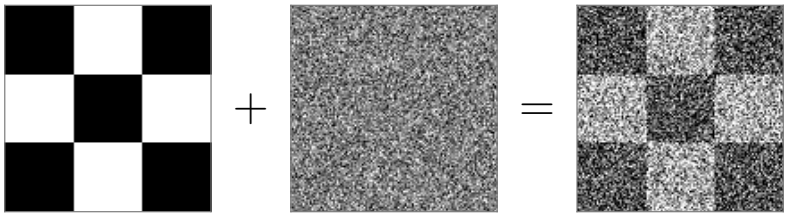
here, about 24°

a rotation of that size can misassign only a small share of the 256 goods
in this draw none are misassigned; sharper entrywise results explain why

backup

recovery from low-rank structure

recovery is possible when the signal has **low-rank structure** absent from the noise

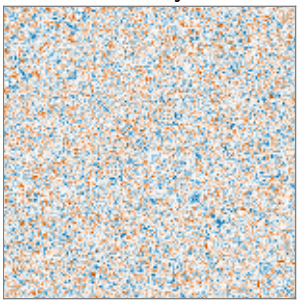


original image | noise | corrupted image

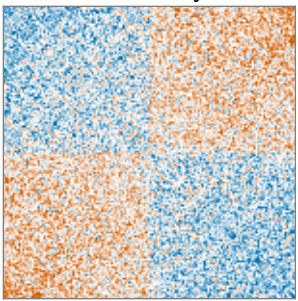
the original has low rank: a few singular components suffice;
the noise has no low-rank structure

ordering the noisy demand matrix $\widehat{\boldsymbol{D}}$

$\widehat{\boldsymbol{D}}$, arbitrary order

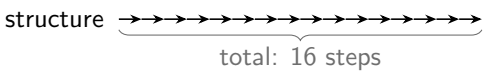


$\widehat{\boldsymbol{D}}$, sorted by $\widehat{\boldsymbol{u}}^1$



- the authority does not observe the goods' group assignments
- sorting goods by the top eigenvector of $\widehat{\boldsymbol{D}}$ recovers the blocks
- individual entries are noisy; block structure is recoverable

signal and noise along eigenvectors



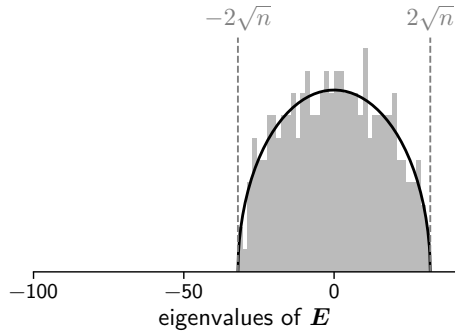
along the top eigenvector $\boldsymbol{u}^1$, demand effects reinforce

$$|(\boldsymbol{u}^1)^\top \boldsymbol{D}\, \boldsymbol{u}^1| = |\lambda_1| \approx 81$$



along a fixed unit vector $\boldsymbol{u}$, random signs produce cancellation

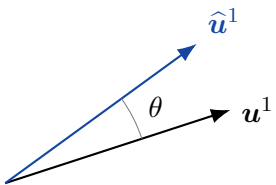
$$\boldsymbol{u}^\top \boldsymbol{E}\, \boldsymbol{u} = O(1)$$

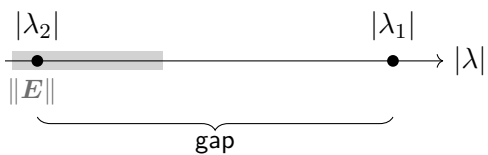


the largest noise eigenvalue is of order $\sqrt{n}$: semicircle law Wigner (1955)

$$\|\boldsymbol{E}\| = \max_{\|\boldsymbol{u}\|=1} |\boldsymbol{u}^\top \boldsymbol{E}\, \boldsymbol{u}| \approx 2\sqrt{n}$$

eigenvector stability



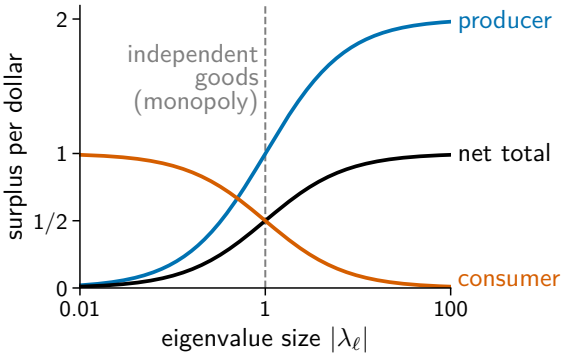


- **Davis–Kahan** (1970): the top eigenvector of $\widehat{\boldsymbol{D}}$ is close to that of $\boldsymbol{D}$ when the eigengap is large relative to the noise norm

$$\sin\theta \lesssim \frac{\|\boldsymbol{E}\|}{|\lambda_1| - |\lambda_2|}$$

- noise norm of order $\sqrt{n}$, eigengap 80: bound $25°$; realized angle $12°$
- without an internal eigengap, the top eigenspace is recovered (B18)

incidence by eigenvalue



- per \$1 along an eigenvector of size $|\lambda_\ell|$, consumers receive $\frac{1}{1+|\lambda_\ell|}$: from \$1 to \$0 as $|\lambda_\ell|$ increases
- producers receive $\frac{2|\lambda_\ell|}{1+|\lambda_\ell|}$: from \$0 to \$2; market surplus $\frac{1+2|\lambda_\ell|}{1+|\lambda_\ell|}$ rises from \$1 to \$2
- eigenvalue 1 (a single firm): 50¢, \$1, 50¢
- large eigenvalues: 33, 81, 500 give \$1.97, \$1.99, \$1.998 of market surplus

the two-ecosystem experiment

$n = 256$ goods, two rival ecosystems: complements within, substitutes across

$$\boldsymbol{D} = -|\lambda_1|\,\boldsymbol{u}^1(\boldsymbol{u}^1)^\top + \underbrace{\text{aggregate}}_{|\lambda|=0.01} + \underbrace{\text{bulk}}_{|\lambda|\approx 1}$$

top price direction $\boldsymbol{u}^1$: "+A, −B";
$|\lambda_1| \in [3, 240]$

- quantities mainly load on the aggregate eigenvector; subsidies mainly lower prices (output response $\approx 1\%$)
- noise matrix $\boldsymbol{E}$: symmetric, i.i.d. $N(0,1)$ entries; $\|\boldsymbol{E}\|$ of order $\sqrt{n}$
- rule: subsidize along estimated direction $\widehat{\boldsymbol{u}}^1$, scaled by observed quantities $\widehat{\boldsymbol{q}}^0$
- 500 draws for each $|\lambda_1|$

the two-ecosystem corollary, formally

an unknown split of the n goods into halves A, B;
$D_{ii} = -1$, $D_{ij} = -c$ within, $D_{ij} = s \geq 0$ across;
$q_i^0 \in [q_{\min}, q_{\max}]$, $q_{\min} > 0$;
noise entries independent, mean zero,
bounded fourth moments, variances bounded below

Corollary. Let Θ contain every such market with $c \geq c_n$ and either
(i) $s \leq \theta c$ for a fixed $\theta < 1$, or
(ii) $|Q_A - Q_B| \geq \delta_0(Q_A + Q_B)$.

If $c_n\sqrt{n} \to \infty$, significant structure holds with $b(n) = (1-\theta)\, c_n\, n/2$, and the conclusions of Theorem 1 hold ε-robustly on Θ.

If $c_n\sqrt{n}$ does not diverge, significant structure fails for every $b(n)$.

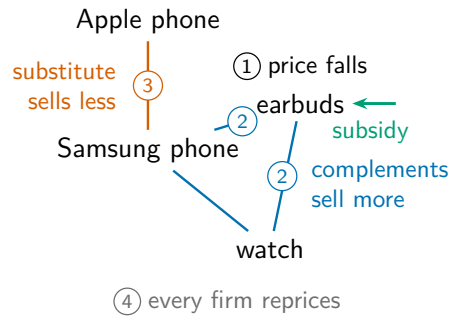
- eigenvalues of $\boldsymbol{D}$:
  $-[(1-c) + \frac{n}{2}(c+s)]$ on $\mathbf{1}_A - \mathbf{1}_B$,
  $-[(1-c) + \frac{n}{2}(c-s)]$ on $\mathbf{1}$,
  $-(1-c)$ on the rest;
  negative semidefinite iff $c \leq 1$, $s \leq c + \frac{1-c}{n/2}$
- noise $\|\boldsymbol{E}\| \approx 2\sqrt{n}$
- the film: $s \approx c$, case (ii); the rule subsidizes the larger-selling ecosystem and taxes the other

equilibrium price responses

1. a subsidy on earbuds lowers their price
2. complements (phone, watch) sell more
3. the substitute (the rival phone) sells less
4. all firms adjust their prices

equilibrium changes in prices $\Delta \boldsymbol{p}$ and quantities $\Delta \boldsymbol{q}$, given subsidy $\boldsymbol{s}$ and demand matrix $\boldsymbol{D}$:

$$\Delta \boldsymbol{p} = -(\boldsymbol{I} - \boldsymbol{D})^{-1}\boldsymbol{s}, \qquad \Delta \boldsymbol{q} = \boldsymbol{D}\,\Delta \boldsymbol{p}$$



households and firms

household h chooses quantities $\boldsymbol{q}^h$ at prices $\boldsymbol{p}$:

$$\max_{\boldsymbol{q}^h} \; U^h(\boldsymbol{q}^h) - \boldsymbol{p} \cdot \boldsymbol{q}^h$$

U^h arbitrary: complements, substitutes

aggregate demand

$$\boldsymbol{q}(\boldsymbol{p}) = \textstyle\sum_h \boldsymbol{q}^h(\boldsymbol{p})$$

n^2 own- and cross-price effects

firm i chooses price p_i, given per-unit subsidy s_i

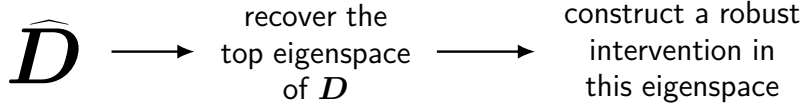
$$\max_{p_i} \; q_i(\boldsymbol{p}) \left(p_i - c_i + s_i\right)$$

$c_i - s_i =$ effective marginal cost; $s_i < 0$ is a tax

equilibrium: pure-strategy Nash $(\boldsymbol{p}, \boldsymbol{q}(\boldsymbol{p}))$, every price a best response; status quo $(\boldsymbol{p}^0, \boldsymbol{q}^0)$ at $\boldsymbol{s} = \boldsymbol{0}$

Bertrand: firms set prices; one good per firm

spectral price theory



theorem (informal): if the top eigenvalue of D is known to be much larger than the noise norm, ε-robust interventions exist

$$|\lambda_1| \gg \|E\|$$

assumptions and significant structure

maintained assumptions

1. the equilibrium is locally unique
2. each good's demand is locally linear in its own price
3. own-price slopes are bounded
4. own-price slopes can be estimated from noisy demand $\widehat{\boldsymbol{D}}$

significant structure, formally: a threshold $b(n) \to \infty$ with

- $\|\boldsymbol{E}\|/b(n) \to 0$: noise is negligible relative to $b(n)$
- eigenvectors of $\boldsymbol{D}$ with $|\lambda| \geq b(n)$ span $\mathcal{L}$
- status-quo quantities $\boldsymbol{q}^0$ load on $\mathcal{L}$: $\|\mathbb{P}_{\mathcal{L}}\boldsymbol{q}^0\| \geq \delta \|\boldsymbol{q}^0\|$
- quantity noise no larger than quantities

← back to the theorem

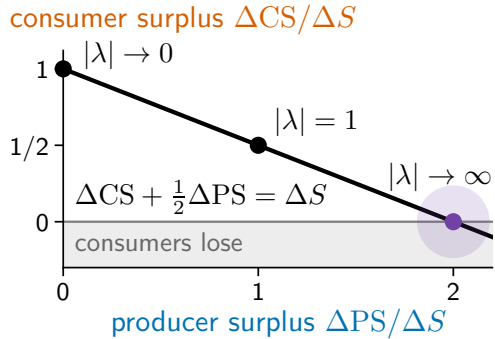
per-dollar incidence

per \$1 on eigenvector ℓ	pass-through $\frac{1}{1+\lvert\lambda_\ell\rvert}$	consumers $\frac{1}{1+\lvert\lambda_\ell\rvert}$	producers $\frac{2\lvert\lambda_\ell\rvert}{1+\lvert\lambda_\ell\rvert}$	market surplus $\frac{1+2\lvert\lambda_\ell\rvert}{1+\lvert\lambda_\ell\rvert}$
small ($\lvert\lambda\rvert = 0.1$)	91%	91¢	18¢	\$1.09
one firm ($\lvert\lambda\rvert = 1$)	50%	50¢	\$1	\$1.50
top eigenvector ($\lvert\lambda_1\rvert = 81$)	1%	1¢	\$1.98	\$1.99

- $\Delta\mathrm{CS} + \frac{1}{2}\Delta\mathrm{PS} = \Delta S$ for each eigenvector
- the split depends on the eigenvalue $\lvert\lambda_\ell\rvert$ alone

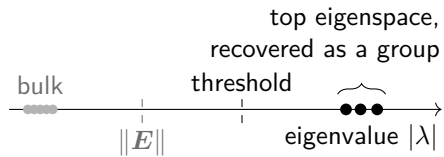
attaining the incidence frontier

- if goods interact ($D_{ij} \neq 0$), demand matrix $\boldsymbol{D} \neq -\boldsymbol{I}$: two eigenvectors with different eigenvalues $|\lambda_k| \neq |\lambda_\ell|$
- each eigenvector yields consumer gain $1/(1+|\lambda_\ell|)$ per dollar
- divide spending between the two (taxing one if needed) to vary consumer gains continuously
- every point on the line is attainable (Proposition 1, part 2)



recovery without an internal eigengap

- clustered large eigenvalues prevent recovery of individual eigenvectors; the top eigenspace remains recoverable
- nonnegative spending on each eigenvector of this eigenspace yields about \$2 to producers and \$0 to consumers per dollar; weighting by observed purchases ensures nonnegative spending
- the rule retains eigenvectors with estimated eigenvalue $|\hat{\lambda}|$ above three quarters of the threshold



demand estimation error

- for each pair of goods, the authority samples one household and runs an unbiased two-good demand experiment, independent across pairs
- repeated estimation of own-price slopes yields precise estimates
- for n goods, $\|\boldsymbol{E}\| \approx 2\sqrt{n}$; weak correlation across experiments preserves this order

guarantees in expectation

Theorem 1

- holds with high probability in every market consistent with the authority's information
- holds in expectation in every such market after capping the rule on rare extreme draws

Proposition 3, part 1

- for every rule spending \$1, expected net gain is negligible in some market in the stated class
- guarantees are prior-free and hold market by market Manski (2004)

mathematical details

impact on equilibrium

$$\boldsymbol{D} = \begin{pmatrix} D_{11} & \cdots & D_{1n} \\ \vdots & \ddots & \vdots \\ D_{n1} & \cdots & D_{nn} \end{pmatrix}$$

$$D_{ij} = \frac{\partial q_i}{\partial p_j}(\boldsymbol{p}^0)$$

effect of p_j on demand for i

$D_{ij} > 0$: j is a **substitute** for i
$D_{ij} < 0$: j is a **complement** to i

- **status quo**: $\boldsymbol{s} = \boldsymbol{0}$, equilibrium $(\boldsymbol{p}^0, \boldsymbol{q}^0)$
- $\boldsymbol{D}$ is symmetric and negative semidefinite (households maximize utility)
- **assumption**: each good's demand is locally linear in its own price
- **units**: $D_{ii} = -1$
- **result**: first-order equilibrium changes:

$$\Delta\boldsymbol{p} = -(\boldsymbol{I} - \boldsymbol{D})^{-1}\boldsymbol{s}, \qquad \Delta\boldsymbol{q} = \boldsymbol{D}\,\Delta\boldsymbol{p}$$

(Δ denotes the first-order change)

← back

complete information

Proposition 1 (complete information):

1. for every intervention $\boldsymbol{s}$:

$$\Delta\mathrm{CS} + \frac{1}{2}\,\Delta\mathrm{PS} = \Delta S$$

hence

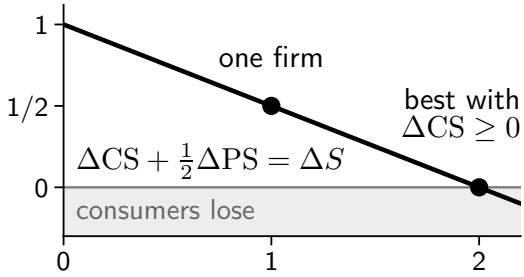
$$\Delta\mathrm{CS} \geq 0 \;\Rightarrow\; \Delta\mathrm{CS} + \Delta\mathrm{PS} \leq 2\,\Delta S$$

2. if goods interact (some $D_{ij} \neq 0$, $i \neq j$), generically every point on the line is attainable

identity:
$\Delta\mathrm{CS} + \frac{1}{2}\Delta\mathrm{PS} = \boldsymbol{q}^0 \cdot (\Delta\boldsymbol{q} - \Delta\boldsymbol{p}) = \boldsymbol{q}^0 \cdot \boldsymbol{s} = \Delta S$

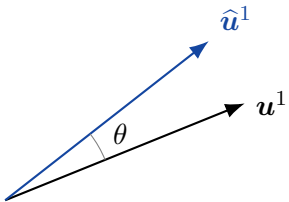
← back

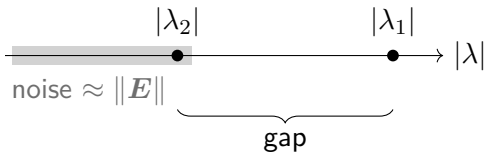
consumer surplus $\Delta\mathrm{CS}/\Delta S$



With $\Delta\mathrm{CS} \geq 0$: at most \$2 of market surplus $\Delta\mathrm{CS} + \Delta\mathrm{PS}$ per \$1 spent, i.e. $\Delta W \leq \Delta S$.

eigenvector recovery





- **Davis–Kahan** (1970): the angle θ between the top eigenvectors of $\widehat{\boldsymbol{D}}$ and $\boldsymbol{D}$ satisfies

$$\sin\theta \lesssim \frac{\|\boldsymbol{E}\|}{|\lambda_1| - |\lambda_2|}$$

- a large gap relative to $\|\boldsymbol{E}\|$ permits precise recovery of the top eigenvector
- the paper recovers the top eigenspace without a gap between its eigenvalues

← back

spectral pass-through

decompose the subsidy into eigenvectors:

$$\boldsymbol{s} \;=\; \sum_{\ell} (\boldsymbol{u}^{\ell} \cdot \boldsymbol{s})\, \boldsymbol{u}^{\ell}$$

Lemma 1 (spectral pass-through):

$$\boldsymbol{u}^{\ell} \cdot \Delta \boldsymbol{p} = -\frac{1}{1+|\lambda_{\ell}|}\, \boldsymbol{u}^{\ell} \cdot \boldsymbol{s}$$

$$\boldsymbol{u}^{\ell} \cdot \Delta \boldsymbol{q} = \frac{|\lambda_{\ell}|}{1+|\lambda_{\ell}|}\, \boldsymbol{u}^{\ell} \cdot \boldsymbol{s}$$

- a subsidy along eigenvector ℓ changes only its price and quantity components
- n scalar coefficients replace the matrix inverse
- demand slope along eigenvector ℓ: $|\lambda_{\ell}|$
- large $|\lambda_{\ell}|$: the subsidy mainly increases **output**
- small $|\lambda_{\ell}|$: the subsidy mainly **lowers prices**

← back

surplus by eigenvector

spending along eigenvector ℓ:

$$\Delta S_\ell = (\boldsymbol{u}^\ell \cdot \boldsymbol{q}^0)(\boldsymbol{u}^\ell \cdot \boldsymbol{s}), \qquad \Delta S = \sum_\ell \Delta S_\ell$$

- the quantity weight is $\boldsymbol{u}^\ell \cdot \boldsymbol{q}^0$, the projection of status-quo quantities

Lemma 2 (surplus by bundle):

$$\Delta \mathrm{CS} = \sum_\ell \frac{1}{1+|\lambda_\ell|}\,\Delta S_\ell$$

$$\Delta \mathrm{PS} = \sum_\ell \frac{2|\lambda_\ell|}{1+|\lambda_\ell|}\,\Delta S_\ell$$

$$\Delta W = \sum_\ell \frac{|\lambda_\ell|}{1+|\lambda_\ell|}\,\Delta S_\ell$$

- surplus per dollar depends only on $|\lambda_\ell|$
- $\Delta \mathrm{CS} + \frac{1}{2}\Delta \mathrm{PS} = \Delta S$ holds for each eigenvector

← back

recovery and welfare

statistics (recovery)

- $|\lambda_\ell| \gg \|\boldsymbol{E}\|$: the associated eigenspace is recoverable
- small $|\lambda_\ell|$: obscured by noise

economics (spectral price theory)

- large $|\lambda_\ell|$: per \$1, $\approx \$2$ to producers, $\approx \$0$ to consumers: $\approx \$2$ of market surplus
- small $|\lambda_\ell|$: larger consumer gains

The recoverable eigenspace yields the highest surplus per dollar.

← back

main result

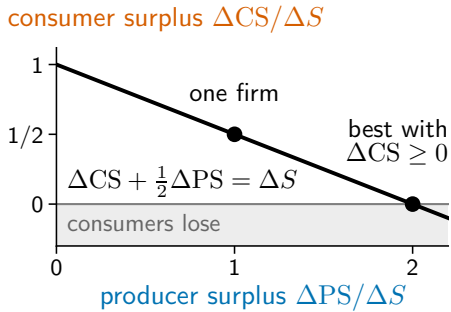
Theorem 1: Suppose the market has significant structure. For every $\varepsilon > 0$ and target spending $\bar{S} > 0$, for large n, there exists a rule that ε-robustly achieves:

1. $\Delta\mathrm{CS} + \Delta\mathrm{PS} \geq (2 - \varepsilon)\,\Delta S$
2. $|\Delta\mathrm{CS}^h| < \varepsilon$ for every household h
3. $|\Delta S - \bar{S}| < \varepsilon$

- $\approx \$2$ of market surplus per dollar ($\Delta W \approx \Delta S$): the complete-information bound with $\Delta\mathrm{CS} \geq 0$
- producer gains can be redistributed through fees and rebates

(regularity conditions: B15)

← back



the intervention rule

1. let $\widehat{\mathcal{L}}$ span the eigenvectors of $\widehat{\boldsymbol{D}}$ whose eigenvalue magnitudes are far above the noise level
2. project the observed quantities $\widehat{\boldsymbol{q}}^0$ onto $\widehat{\mathcal{L}}$
3. scale the projection to target spending $\bar{S}$:

$$\boldsymbol{s} \propto \mathbb{P}_{\widehat{\mathcal{L}}}\, \widehat{\boldsymbol{q}}^0$$

one top eigenvector:

$$\boldsymbol{s} = \frac{\bar{S}\, \widehat{\boldsymbol{u}}^1}{\widehat{\boldsymbol{u}}^1 \cdot \widehat{\boldsymbol{q}}^0}$$

- uses eigenspaces rather than matrix inversion
- quantities determine sign and scale; eigenvectors $\boldsymbol{u}^\ell$ and $-\boldsymbol{u}^\ell$ are equivalent
- spending follows projected quantities

← back

the limits of robust interventions

Proposition 3 (limits): Let $\Delta S > 0$.

1. **without** significant structure: in some markets, no rule ε-robustly achieves

$$\Delta\mathrm{CS} + \Delta\mathrm{PS} \;\geq\; (1+\varepsilon)\,\Delta S$$

   no positive net return is guaranteed, even on average

2. **even with** it: in some such markets, no rule robustly gives consumers a fixed share

$$\Delta\mathrm{CS} \;\geq\; \kappa\,\Delta S \qquad (\text{any } \kappa > 0)$$

- (1) noise prevents identification of profitable directions

- (2) consumer gains require small-$|\lambda_\ell|$ directions, obscured by noise

← back